

NUMERICAL METHODS: LABORATORY REPORT 03

Topic: Real-World Data Linear Regression (DPWH Infrastructure Data)

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PART A. Data

Source & Full URL: Department of Public Works and Highways (DPWH) Regional Infrastructure Summary
URL: <https://www.dpwh.gov.ph/dpwh/gis/stat>
Description: Summary of national road and bridge counts and total lengths (km) across administrative regions in the Philippines.
Variables: Independent (x) = Total Infrastructure Count | Dependent (y) = Total Length (km)

Region	Total Count (x)	Total Length in km (y)
CAR	355	14,836.64
NCR	307	30,932.05
NIR	437	17,002.91
Region I	591	36,586.86
Region II	522	30,479.54
Region III	759	39,832.49
Region IV-A	695	21,096.84
Region IV-B	661	24,990.23
Region V	738	25,007.43
Region VI	529	26,689.43
Region VII	435	18,873.18
Region VIII	928	37,492.87
Region IX	393	16,854.75
Region X	469	21,040.71
Region XI	345	17,932.10
Region XII	326	13,230.94
Region XIII	466	22,533.77
BARMM	175	6,978.95

PART B. Python Script

```
import inspect
import os
import matplotlib.pyplot as plt
import numpy as np
from reportlab.lib import colors
from reportlab.lib.pagesizes import letter
from reportlab.lib.styles import ParagraphStyle, getSampleStyleSheet
from reportlab.platypus import (
    HRFflowable,
    Image,
    Paragraph,
    Preformatted,
    SimpleDocTemplate,
    Spacer,
    Table,
    TableStyle,
)

# -----
# 1. DATASET & LEAST SQUARES CALCULATIONS
# -----

regions = [
    "CAR",
    "NCR",
    "NIR",
    "Region I",
    "Region II",
    "Region III",
    "Region IV-A",
    "Region IV-B",
    "Region V",
    "Region VI",
    "Region VII",
    "Region VIII",
    "Region IX",
    "Region X",
    "Region XI",
    "Region XII",
```

```

"Region XIII",
"BARM",
]

# x = Total Infrastructure Segments / Bridges / Roads (Count)
x = np.array(
    [
        355,
        307,
        437,
        591,
        522,
        759,
        695,
        661,
        738,
        529,
        435,
        928,
        393,
        469,
        345,
        326,
        466,
        175,
    ],
    dtype=float,
)

# y = Total Infrastructure Length (km)
y = np.array(
    [
        14836.64,
        30932.05,
        17002.91,
        36586.
    ]

# ... [Script Code Truncated for Page Layout] ...

```

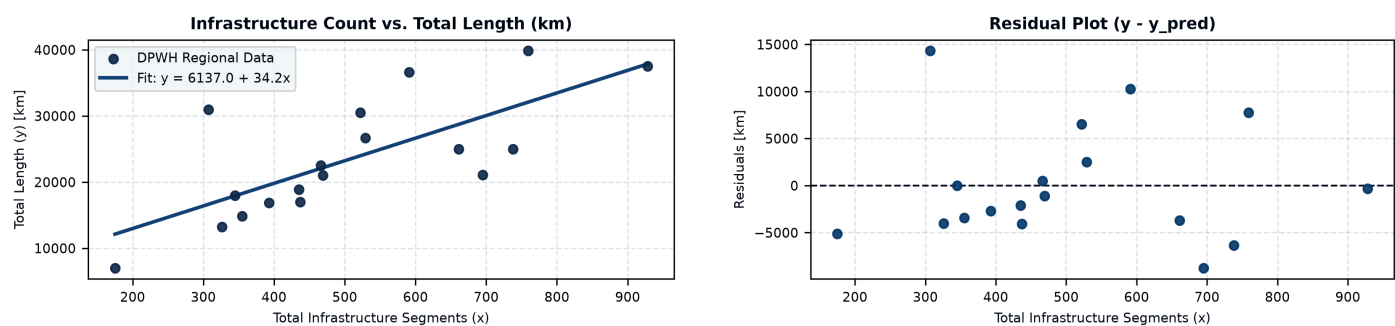
PART C. Regression Results

01	02	03	04
MODEL FIT	ACCURACY	ERROR	PREDICTION
y = 6137.0 + 34.2x	r² = 0.5326	s _{y/x} = 6,297 km	23,217.6 km

1. Regression Equation & Calculated Metrics

Metric	Formula / Notation	Calculated Value
Fitted Regression Model	y = a0 + a1 * x	y = 6137.0402 + 34.1611x
Intercept (a0)	a0 = y_bar - a1 * x_bar	6,137.0402 km
Slope (a1)	a1 = (n*Σxy - ΣxΣy) / (n*Σx² - (Σx)²)	34.1611 km / segment
Sum of Squared Residuals (SSE)	Sr = Σ(y - y_pred)²	634,494,200.26
Coefficient of Determination	r² = (St - Sr) / St	0.5326
Standard Error of Estimate	s_y/x = sqrt(Sr / (n - 2))	6,297.2921 km

2. Graph with Fitted Line & Residual Plot



3. Interpretation of Results

The simple linear regression model reveals a moderate positive linear correlation between total infrastructure counts and total network lengths across DPWH administrative regions.

- Slope (a1 = 34.1611):** Each additional infrastructure segment adds approximately **34.16 km** of road/bridge network length.
- Intercept (a0 = 6,137.0402):** Represents the baseline regional network span offset when segment count is zero.
- Coefficient of Determination (r² = 0.5326):** **53.26%** of total regional length variance is explained by the number of segments.

- **Standard Error ($s_{y/x} = 6,297.2921$):** Regional predictions deviate by an average of **6,297.29 km**.
- **Residuals Analysis:** Shows higher variance in regions like NCR (high density) and BARMM, indicating varying average segment lengths across regions.

4. Out-of-Sample Prediction

Target Segment Count (x): 500 segments | **Predicted Length (y):** 23,217.59 km
Substitution: $y = 6137.0402 + (34.1611 * 500)$
Significance: Provides baseline linear forecasting for budgeting and scaling regional infrastructure expansion.